

STEP-BY-STEP GUIDE · ROCKY LINUX 9 · JUNE 2026

CogVideo on Your RTX 3090

Type a sentence. Get a video clip. This guide installs everything you need to run CogVideo completely locally — no cloud, no subscription. Estimated time: 30–45 minutes, mostly waiting for downloads.

YOUR MACHINE — CHECKED

GPU

RTX 3090

VRAM

24 GB

RAM

30 GB

DISK FREE

832 GB

PYTHON

Fix in step 1

Your GPU is not "old" for this — an RTX 3090 with 24 GB VRAM runs every CogVideo model without strain. The only thing to fix is Python (you have 3.9; CogVideo needs 3.11). Step 1 handles that.

01

Install Python 3.11

CogVideo requires Python 3.10–3.12. Your system has 3.9. Run this once — it takes about a minute.

TERMINAL

```
sudo dnf install python3.11 python3.11-pip -y
```

When it finishes, confirm it worked:

VERIFY

```
python3.11 --version
```

EXPECTED OUTPUT

```
Python 3.11.x
```

02

Download the CogVideo Code

This copies the project files to your computer. The second command moves you into that folder.

TERMINAL

```
git clone https://github.com/zai-org/CogVideo.git
cd CogVideo
```

Important: Every command from here on should be run inside this `cogvideo` folder. If you open a new terminal window later, run `cd cogvideo` again before anything else.

03

Create an Isolated Environment

A "virtual environment" keeps CogVideo's packages separate from the rest of your system. Think of it as a dedicated workspace.

TERMINAL

```
python3.11 -m venv venv
source venv/bin/activate
```

WHAT TO LOOK FOR

Your terminal prompt now starts with `(venv)` – that tells you the environment is active.

Each new terminal session: you need to re-activate. Go to the CogVideo folder and run `source venv/bin/activate` again. You won't need to reinstall anything — just that one command.

04

Install PyTorch with GPU Support

PyTorch is the AI engine underneath CogVideo. We install a build that speaks directly to your RTX 3090 via CUDA.

Make sure your prompt shows `(venv)` before running this. The download is about 2–3 GB.

TERMINAL

```
pip install torch torchvision \
  --index-url https://download.pytorch.org/whl/cu128
```

Once it finishes, run this one-liner to confirm your GPU is visible:

VERIFY

```
python -c "import torch; print('GPU found:', torch.cuda.get_device_name(0))"
```

EXPECTED OUTPUT

```
GPU found: NVIDIA GeForce RTX 3090
```

If the output says **False** or throws an error, try replacing `cu128` with `cu124` in the install command and run it again.

05

Install CogVideo's Packages

This installs all the remaining libraries CogVideo depends on. Lots of text scrolls by — that's normal.

TERMINAL

```
pip install -r requirements.txt
```

Takes 3–6 minutes depending on your connection. It's done when your `(venv)` prompt comes back and you're able to type again.

06

Connect Your HuggingFace Account

CogVideo's AI model is hosted on HuggingFace (a free platform). You need an account to download it.

6a. Create a free account at huggingface.co if you don't have one.

6b. On HuggingFace, search for [THUDM/CogVideoX-5b](#) and open the model page. Click *"Agree and access repository"* to accept the license (a one-time click).

6c. Go to your HuggingFace account → Settings → Access Tokens → create a new token with *Read* access. Copy the token string.

6d. Back in your terminal, run:

TERMINAL

```
huggingface-cli login
```

It will ask for your token. Paste it and press Enter. You won't see the characters as you type — that's expected for passwords.

EXPECTED OUTPUT

```
Login successful
```

07

Generate Your First Video

The model downloads automatically on the first run (~17 GB — a few minutes). After that, generation starts immediately.

Run this from inside the [CogVideo](#) folder with your `venv` active:

TERMINAL

```
python inference/cli_demo.py \  
  --prompt "A red kite soaring over green valleys at golden hour" \  
  --model_path THUDM/CogVideoX-5b \  
  --generate_type t2v \  
  --dtype bfloat16 \  
  --output_path ./my_first_video.mp4
```

Change the `--prompt` text to anything you want to generate. Your video saves as `my_first_video.mp4` inside the CogVideo folder.

With your RTX 3090, generation takes roughly **3–8 minutes** for a typical prompt. A progress bar appears in the terminal while it works.

WHEN IT FINISHES

100%|██████████| 50/50 [04:12<00:00, ...] Video saved to ./my_first_video.mp4

If something goes wrong

? The GPU verify step prints False or crashes

The CUDA version in the PyTorch URL might not match your driver. Re-install PyTorch with an older CUDA build:

```
pip install torch torchvision --index-url https://download.pytorch.org/whl/
```

? "python3.11: command not found" after Step 1

The install may not have completed. Run Step 1 again and wait for the [Complete!](#) message at the end. Check your internet connection if it keeps failing.

? I closed the terminal and now commands don't work

You need to re-activate the virtual environment. Open a terminal and run:

```
cd ~/CogVideo
source venv/bin/activate
```

Once you see `(venv)` in your prompt, everything works again. You never need to reinstall.

? "CUDA out of memory" during generation

With 24 GB VRAM this is rare, but if it happens: try a shorter video with `--num_frames 49`, or switch to the lighter 2B model by changing `--model_path THUDM/CogVideoX-2b`.

? pip install says "no matching distribution found"

Check that your venv is active — run `which python` and the path should include `/venv/` in it. If not, activate it first with `source venv/bin/activate`.

? HuggingFace login fails or says "401 Unauthorized" during download

Make sure you clicked *"Agree and access repository"* on the model's HuggingFace page (Step 6b) — the token alone is not enough, you also need to accept the license. Then run `huggingface-cli login` again with a fresh token.

Next steps: once generation works, try `THUDM/CogVideoX1.5-5b` as the model path for higher-resolution output (your 24 GB GPU handles it). The `inference/` folder also contains scripts for image-to-video and video-to-video generation.